

FS example —

Problem where you do not need to apply analysis equation separately —

$$x(t) = 3 + 11 \sin(\omega t + 30^\circ) + 7 \cos 3\omega t + 5 \sin 3\omega t + 3 \sin 7\omega t + \cos 11\omega t$$

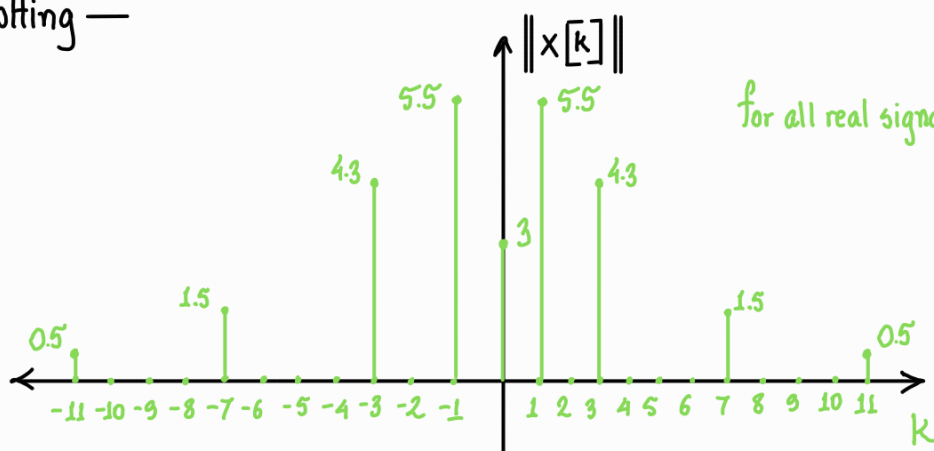
Draw the $X[k\omega]$ for this signal.

$$x(t) = 3 + 11 \sin 30^\circ \cos \omega t + 11 \cos 30^\circ \sin \omega t + \frac{7}{2} e^{j3\omega t} + \frac{7}{2} e^{-j3\omega t} + \frac{5}{2j} e^{j3\omega t} - \frac{5}{2j} e^{-j3\omega t} \\ + \frac{3}{2j} e^{j7\omega t} - \frac{3}{2j} e^{-j7\omega t} + \frac{1}{2} e^{j11\omega t} + \frac{1}{2} e^{-j11\omega t}$$

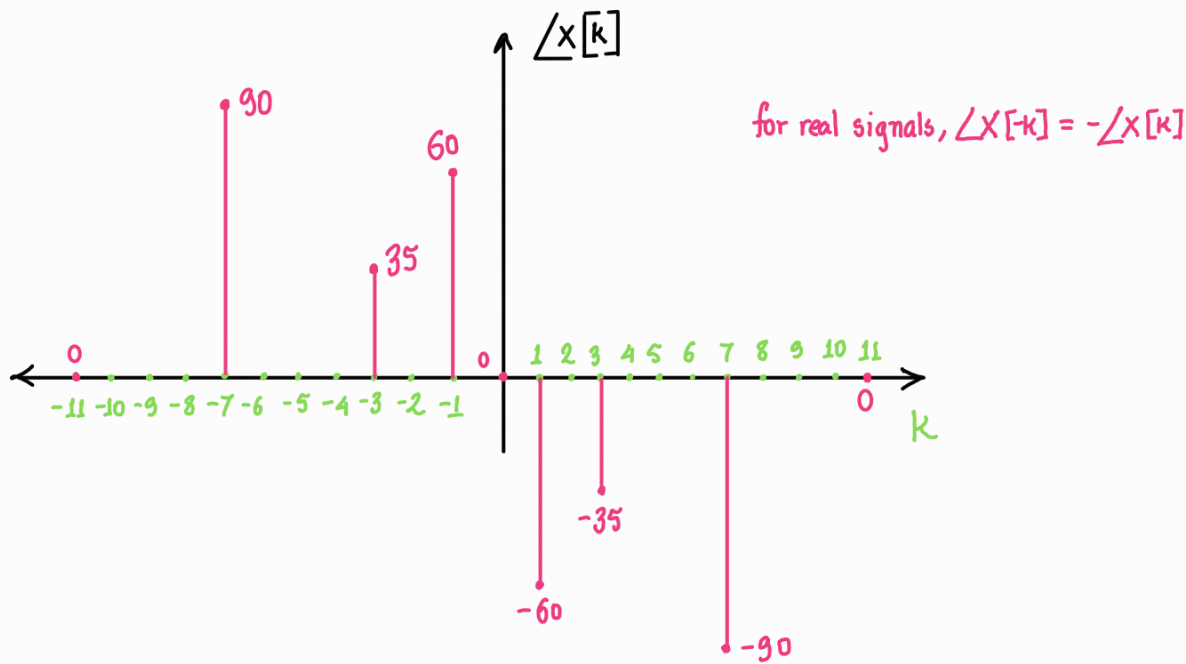
$$= 3 + \frac{5.5}{2} e^{j\omega t} + \frac{5.5}{2} e^{-j\omega t} + \frac{5.5\sqrt{3}}{2j} e^{j\omega t} - \frac{5.5\sqrt{3}}{2j} e^{-j\omega t} + \frac{7j+5}{2j} e^{j3\omega t} + \frac{7j-5}{2j} e^{-j3\omega t} \\ + \frac{3}{2j} e^{j7\omega t} - \frac{3}{2j} e^{-j7\omega t} + \frac{1}{2} e^{j11\omega t} + \frac{1}{2} e^{-j11\omega t}$$

$$= \frac{1}{2} e^{-j11\omega t} - \frac{3}{2j} e^{-j7\omega t} + \frac{7j-5}{2j} e^{-j3\omega t} + \frac{5.5j-5.5\sqrt{3}}{2j} e^{-j\omega t} + 3 + \frac{5.5j+5.5\sqrt{3}}{2j} e^{j\omega t} + \frac{7j+5}{2j} e^{j3\omega t} \\ + \frac{3}{2j} e^{j7\omega t} + \frac{1}{2} e^{j11\omega t}$$

Plotting —



For all real signals $\|X[-k]\| = \|X[k]\|$



One more problem but for DT —

$$x[n] = 3 + 12\sin\left(\frac{3\pi}{7}n\right) + 3\cos\left(\frac{5\pi}{7}n\right) - \sin\left(\frac{9\pi}{7}n\right) + \cos\left(\frac{17\pi}{7}n\right). \text{ Find } x[k]$$

at first we need to figure out Ω_0 . Here, $N=14$ for all components. So, $\Omega_0 = \frac{\pi}{7}$

$$\text{then } x[n] = 3 + 12\sin(3\Omega_0 n) + 3\cos(5\Omega_0 n) - \sin(9\Omega_0 n) + \cos(17\Omega_0 n)$$

$$\text{since } N=14, \cos(17\Omega_0 n) = \cos(9\Omega_0 n)$$

✓ will cause aliasing
max fidelity is π rad/sample
-ve axis is just a reflection

$$\text{so, } x[n] = 3 - 6je^{j3\Omega_0 n} + 6je^{-j3\Omega_0 n} + 1.5e^{j5\Omega_0 n} + 1.5e^{-j5\Omega_0 n} + 0.5je^{j9\Omega_0 n} - 0.5je^{-j9\Omega_0 n} + 0.5e^{j3\Omega_0 n} + 0.5e^{-j3\Omega_0 n}$$

$$= -0.5je^{-j9\Omega_0 n} + 1.5e^{-j5\Omega_0 n} + (0.5+6j)e^{-j3\Omega_0 n} + 3 + (0.5-6j)e^{j3\Omega_0 n} + 1.5e^{j5\Omega_0 n} + 0.5je^{j9\Omega_0 n}$$

Comparing $x[n]$ to standard synthesis equation, we see that there are -ve values of $k\Omega_0$ that needs to be changed to +ve.

$$-9\Omega_0 = (-14+5)\Omega_0 = 5\Omega_0$$

$$-5\Omega_0 = (-14+9)\Omega_0 = 9\Omega_0$$

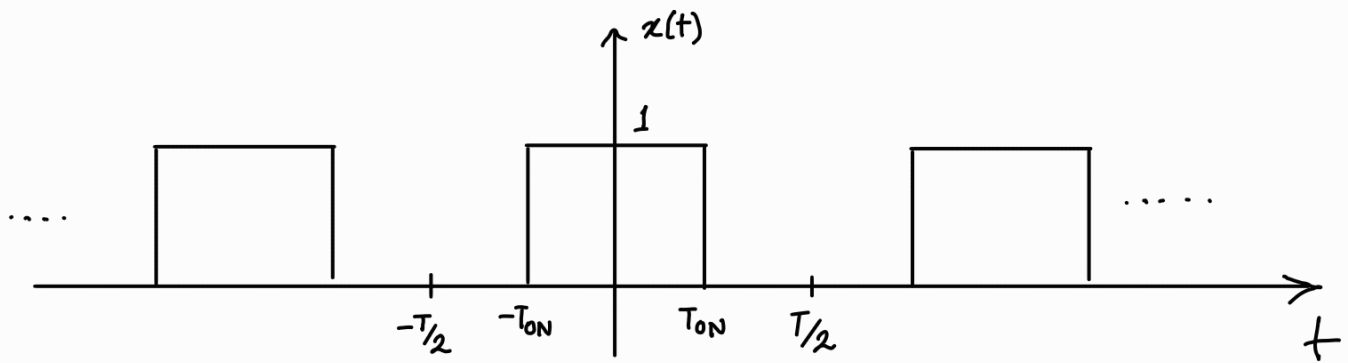
$$-3\Omega_0 = (-14+11)\Omega_0 = 11\Omega_0$$

$$\therefore x[n] = 3 + (0.5-6j)e^{j3\Omega_0 n} + (1.5-0.5j)e^{j5\Omega_0 n} + (1.5+0.5j)e^{j9\Omega_0 n} + (0.5+6j)e^{j11\Omega_0 n}$$

$$x[k] = \left\{ \dots, \underset{\uparrow}{3}, 0, 0, 0.5-6j, 0, 1.5-0.5j, 0, 0, 0, 1.5+0.5j, 0, 0.5+6j, 0, 0, \dots \right\}$$

Complex signals that do require us to apply the formulas —

Find $x[k]$ for



$$x[k] = \frac{1}{T} \int_{-T/2}^{T/2} x(t) \cdot e^{-jk\omega_0 t} dt$$

$$= \frac{1}{T} \int_{-T_{ON}}^{T_{ON}} e^{-jk\omega_0 t} dt$$

$$= \frac{j}{k\omega_0 T} e^{-jk\omega_0 t} \Big|_{-T_{ON}}^{T_{ON}}$$

$$= \frac{j}{k\omega_0 T} \left[e^{-jk\omega_0 T_{ON}} - e^{jk\omega_0 T_{ON}} \right]$$

$$= \frac{2}{k\omega_0 T} \left[\frac{e^{jk\omega_0 T_{ON}} - e^{-jk\omega_0 T_{ON}}}{2j} \right]$$

$$= \frac{2 \sin k\omega_0 T_{ON}}{k\omega_0 T}$$

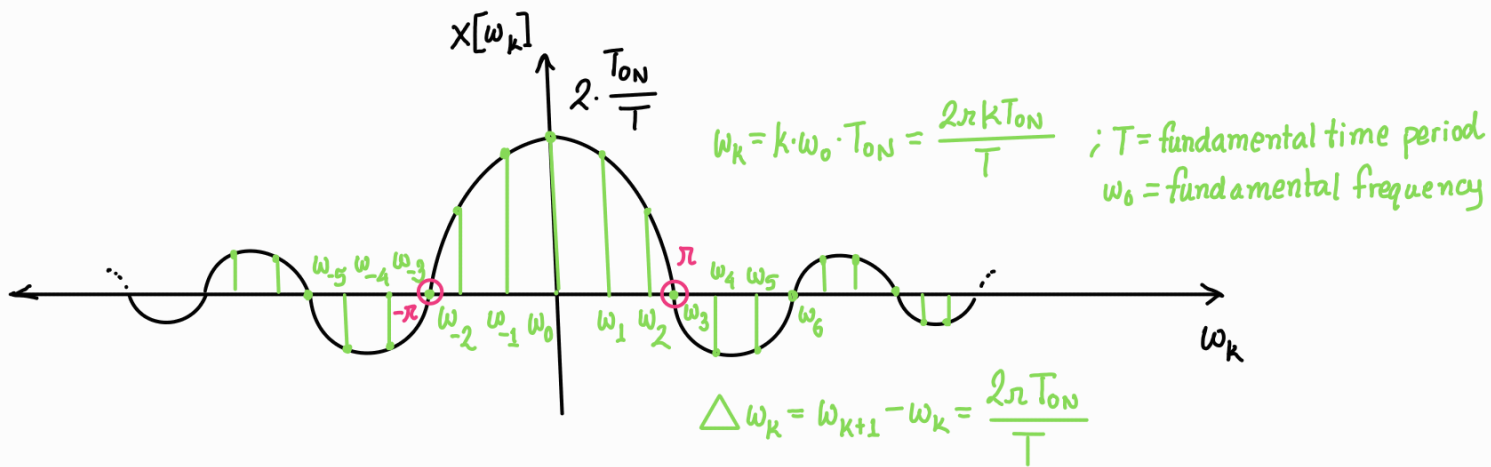
$$= \frac{\sin k\omega_0 T_{ON}}{k\omega_0 T_{ON}} \cdot \frac{2T_{ON}}{T}$$

$$= \frac{2T_{ON}}{T} \operatorname{sinc} k\omega_0 T_{ON}$$

$$D = \text{duty cycle} = \frac{T_{ON}}{T}$$

$$\operatorname{sinc} x = \frac{\sin x}{x}$$

This will look something like —



Building towards something interesting —

Since the spacing between the discrete samples are set by T_{0N} and T value, let's keep T_{0N} fixed and increase T . Then as $\lim T \rightarrow \infty \Delta \omega_k \rightarrow 0$. So the discrete spectrum becomes a continuous spectrum.

But what does this imply in the time domain? Making $\lim T \rightarrow \infty$ makes our square wave aperiodic.

So in theory, although Fourier is primarily designed for periodic signal, we can see a way to extend it for aperiodic signals. We can think of aperiodic signals as periodic signals where $\lim T \rightarrow \infty$.

There is a problem with this though. Look at the amplitude. It has a $\frac{1}{T}$ term which cause it to go towards 0 as $\lim T \rightarrow \infty$. To rectify the issue, we need to figure out the origin of the $\frac{1}{T}$ term.

$$X[k] = \left(\frac{1}{T} \right) \int_{-T/2}^{T/2} x(t) \cdot e^{-jk\omega_0 t} dt$$

origin of the $\frac{1}{T}$ term

The $\frac{1}{T}$ term mainly scales the contributions. So we can formulate an unscaled version —

$$X'[k\omega_0] = \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt$$

$$\text{as } \lim_{T \rightarrow \infty} X'[k\omega_0] = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt$$

$$\Rightarrow X'(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad \left[\because \text{as } \lim T \rightarrow \infty, \lim \omega_0 \rightarrow 0, k\omega_0 \rightarrow \omega \right]$$

discrete $\rightarrow$ continuous

new analysis equation unlocked

We also have to formulate the synthesis equation —

$$x(t) = \sum_{k=-\infty}^{\infty} X[k\omega_0] e^{jk\omega_0 t}$$

$$\begin{aligned} \text{for } \lim_{T \rightarrow \infty} \quad x(t) &= \lim_{T \rightarrow \infty} \sum_{k=-\infty}^{\infty} \left[\frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt \right] e^{jk\omega_0 t} \\ &= \lim_{\omega_0 \rightarrow 0} \sum_{k=-\infty}^{\infty} \frac{\omega_0}{2\pi} X'(k\omega_0) e^{jk\omega_0 t} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X'(j\omega) e^{j\omega t} d\omega \quad \leftarrow \text{new synthesis equation unlocked} \end{aligned}$$

This system of equations are called Fourier Transform (FT)

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

The DTFT can be similarly formulated as —

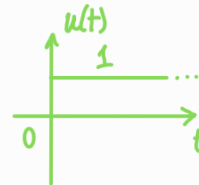
$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega \quad \rightarrow \text{max distinct frequency values from } -\pi \text{ to } +\pi$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

Examples —

Apply CTFT on $x(t) = e^{-t} \cdot u(t)$

$\rightarrow$ unit step function $u(t) = 1 \cdot (t \geq 0)$



$$X(\omega) = \int_{-\infty}^{\infty} e^{-t} u(t) \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-j\omega t - t} dt$$

$$= \frac{1}{-1-j\omega} e^{-t(1+j\omega)} \Big|_0^{\infty}$$

$$= \frac{1}{1+j\omega}$$

You cannot apply CTFT on periodic signals as integrating periodic signals over $-\infty$ to $+\infty$ gives you an undefined sum. But CTFT can exist for periodic signals too. We have to use unit impulse responses to formulate $X(\omega)$ taking the help of FS. We generally work backwards.

Apply ICTFT on $2\pi \delta(\omega - \omega_0)$

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega t} d\omega \\ &= e^{j\omega_0 t} \end{aligned}$$

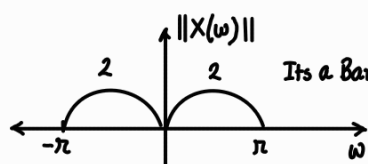
So $\text{CTFT}(e^{j\omega_0 t}) = 2\pi \delta(\omega - \omega_0) \leftarrow \text{Cannot be extracted directly because } e^{j\omega_0 t} \text{ is periodic.}$

Question — Prove $x(t) = \cos \omega_0 t \iff X(\omega) = \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$

$$\begin{aligned} \text{FT}(\cos \omega_0 t) &= \frac{1}{2} \text{FT}(e^{j\omega_0 t} + e^{-j\omega_0 t}) \\ &= \frac{1}{2} (2\pi \delta(\omega - \omega_0) + 2\pi \delta(\omega + \omega_0)) \\ &= \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0) \end{aligned}$$

Apply DTFT on $x[n] = \{-1, 0, 1\}$

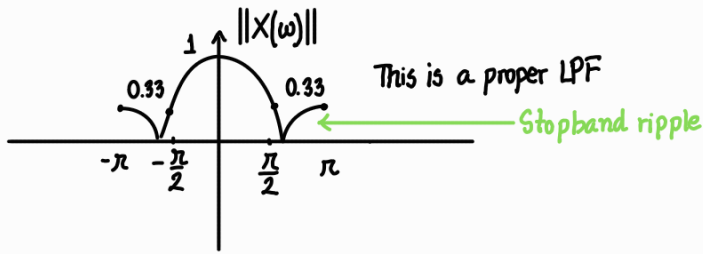
$$X(\omega) = e^{-j\omega} - e^{j\omega} = -2j \sin \omega$$



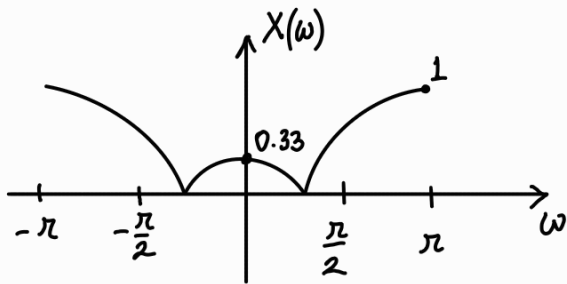
Its a Bandpass filter, not a HP filter as we assumed before

Apply DTFT on $x[n] = \{1, \underset{\uparrow}{1}, 1\} \times \frac{1}{3}$

$$X(\omega) = \frac{1}{3}(e^{-j\omega} + 1 + e^{j\omega}) = \frac{1}{3}(2\cos \omega + 1)$$



Apply IDTFT on $X(\omega) = \frac{1}{3}(1 - 2\cos \omega)$



$$\begin{aligned} X(\omega) &= \frac{1}{3}(1 - e^{-j\omega} - e^{j\omega}) \\ &= \frac{1}{3}(-e^{-j\omega} + 1 - e^{j\omega}) \end{aligned}$$

You don't even need to apply IDTFT here. By comparing,

$$x[n] = \left\{ \frac{-1}{3}, \frac{1}{3}, \frac{-1}{3} \right\} \leftarrow \text{this is a HP filter}$$

$$= \frac{1}{3} \times \left\{ \underset{\uparrow}{-1}, 1, -1 \right\}$$

$X(\omega) = \frac{1}{4}(2 - 2\cos(\omega))$ is a better HPF response. $x[n] = \frac{1}{4}\{-1, 2, -1\}$